Guanghui Qin

https://gqin.me/ {first.last}@recursion.com Google Scholar

Education

Johns Hopkins University Maryland, USA
Ph.D. in Computer Science (Topic: Natural Language Processing) *Aug 2019 – Aug 2024*

Peking University Beijing, China
B.S. in Physics & Computer Science *Sep 2015 – Jun 2019*

Experience

Recursion Pharmaceuticals Utah, USA

- *Staff AI Researcher* *(Manager: Hoifung Poon)* *Oct 2026 – Present*
  Decoding biology with AI for drug discovery.

Microsoft Research Washington, USA

- *Senior Researcher* *(Manager: Hoifung Poon)* *Sep 2024 – Oct 2026*
  Researched in **multimodal** methods for **AI in biomed/science**, including: (1) a **vision model** to predict neurological outcomes from **brain MRI**; (2) a **reinforcement learning** framework that achieved state-of-the-art performance in chest X-ray **report generation**; and (3) a benchmark that evaluates **agents**' performance in healthcare with CLI.
- *Research Intern* *(Mentor: Corby Rosset)* *May 2023 – Aug 2023*
  Conducted research on **efficient methods** for long-context **LLMs**. Proposed a method to **compress** the context of LLaMA by up to 20× with minimal performance tradeoffs. The method improved **retrieval-augmented generation (RAG)**.
- *Research Intern* *(Mentor: Anthony Platanios)* *May 2022 – Aug 2022*
  Studied a new research problem in **user action prediction**. Built a dataset from GitHub with 20 million actions and implemented **GNN models** to predict interactions among users and repositories (e.g., mentions, pull requests, and issues).

Johns Hopkins University Maryland, USA

- *Ph.D. Candidate* *(Advisor: Benjamin Van Durme)* *Aug 2019 – Aug 2024*
  Conducted research in **natural language processing** and **machine learning**, focusing on **efficient methods for large language models**, **prompt tuning**, and **semantic parsing**.
- *Visiting Researcher* *(Mentor: Jason Eisner)* *Jul 2018 – Oct 2018*
  Studied **temporal event stream** modeling. Proposed a particle smoothing method for sampling events from a neural **Hawkes process**. The **stochastic process** could interact with a deductive temporal database such as **Datalog**.

Microsoft Research Asia Beijing, China

- *Research Intern* *(Mentors: Jin-Ge Yao & Chin-Yew Lin)* *Nov 2017 – Jun 2018*
  Proposed a Semi-HMM model for grounding natural language to **structured data** for use in **data-to-text generation**.

Skills

- Programming languages: Python, Java, and C/C++. Other languages: Shell, LaTeX, and SQL.
- Natural Language Processing (PhD topic): Transformers, LLMs, information retrieval, and semantic parsing.
- Computer Vision: CNNs, (3D) ViTs, and medical image processing.
- Supervised fine-tuning and reinforcement learning for multimodal LLMs (verl, MS-SWIFT, etc.).
- Distributed training and performance optimization (runtime and memory).
- Experience processing real-world biomedical data, including DICOM/NIfTI files and luminescence images.
- Physics training as undergraduate armed me with critical and scientific thinking.

Academic Service

Served as an *area chair* for NeurIPS (2026). Served as a *reviewer* for NeurIPS (2019–2023 and 2025), ICLR (2019–2021 and 2023–2027), ICML (2020, 2021, and 2026), CVPR (2026), ACL (2021 and 2025), EMNLP (2019–2022 and 2026), NAACL (2024), AAAI (2021), and AKBC (2020).

Honors

- Best Short Paper Award at NAACL — ACL, 2021
- May 4th Scholarship — Peking University, 2016 and 2018
- Silver Medalist — Chinese Physics Olympiad (CPhO), 2014

Selected Publications

- Evaluating the robustness and readiness of large frontier models in health AI applications. Y. Gu*†, J. Fu*, X. Liu, J.M.J. Valanarasu, N.C.F. Codella, R. Tan, Q. Liu, Y. Jin, S. Zhang, J. Wang, R. Wang, L. Song, **G. Qin**, . . . , E. Topol†, H. Poon†, and P. Vozila. In *Nature Medicine*. 2026.
- Masked-Diffusion Autoencoders for 3D Medical Vision Representation Learning. J. Tu*, **G. Qin***, T. Zhao, J.M.J. Valanarasu, S. Zhang, T. Naumann, F. Lam, S. Wang, and H. Poon. In *CVPR*. 2026.
- Scaling medical imaging report generation with multimodal reinforcement learning. Q. Liu*, S. Zhang*, **G. Qin***, . . . , M. Wei†, and H. Poon†. In *Nature Communications* (to appear). 2026.
- HealthAgentBench: A Unified Benchmark Suite of Realistic Agentic Healthcare Environments for Challenging Frontier AI Agents. Q. Liu*, S. Zhang*, **G. Qin***, . . . , and H. Poon. In *arXiv*. 2026.
- X-Reasoner: Towards Generalizable Reasoning Across Modalities and Domains. S. Zhang*, Q. Liu*, **G. Qin***, T. Ossowski, A. Gu, Y. Jin, S. Kiblawi, S. Preston, M. Wei, P. Vozila, T. Naumann, and H. Poon. In *arXiv*. 2025.
- Ras-mediated dynamic and biphasic regulation of cell migration. Y. Lin, E. Parajón, Q. Yuan, S. Ye, **G. Qin**, . . . and P.N. Devreotes†. In *PNAS*. 2025.
- Researchy Questions: A Dataset of Multi-Perspective, Decompositional Questions for Deep Research. C. Rosset, H.-L. Chung, **G. Qin**, E. Chau, Z. Feng, A. Awadallah†, J. Neville†, and N. Rao†. In *ACM SIGIR*. 2025.
- Dodo: Dynamic Contextual Compression for Decoder-only LMs. **G. Qin**, C. Rosset, E. Chau, N. Rao, and B. Van Durme. In *ACL* (oral presentation). 2024.
- Researchy Questions: A Dataset of Multi-Perspective, Decompositional Questions for Deep Research. C. Rosset, H.-L. Chung, **G. Qin**, E. Chau, Z. Feng, A. Awadallah†, J. Neville†, and N. Rao†. In *ACM SIGIR*. 2025.
- Ras suppression potentiates rear actomyosin contractility-driven cell polarization and migration. Y. Lin*, D.S. Pal*†, P. Banerjee, T. Banerjee, **G. Qin**, . . . , and P.N. Devreotes†. In *Nature Cell Biology*. 2024.
- Nugget: Neural Agglomerative Embeddings of Text. **G. Qin** and B. Van Durme. In *ICML*. 2023.
- The NLP Task Effectiveness of Long-Range Transformers. **G. Qin**, Y. Feng, and B. Van Durme. In *EACL* (oral presentation). 2023.
- Learning How to Ask: Querying LMs with Mixtures of Soft Prompts. **G. Qin** and J. Eisner. In *NAACL* (plenary talk). 2021. ***Best Short Paper Award***
- LOME: Large Ontology Multilingual Extraction. P. Xia*, **G. Qin***, S. Vashishtha, Y. Chen, T. Chen, C. May, C. Harman, K. Rawlins, A.S. White, and B. Van Durme. In *EACL*. 2021.
- Neural Datalog Through Time: Informed Temporal Modeling via Logical Specification. H. Mei, **G. Qin**, M. Xu, and J. Eisner. In *ICML* (oral presentation). 2020.
- Imputing Missing Events in Continuous-Time Event Streams. H. Mei, **G. Qin**, and J. Eisner. In *ICML* (oral presentation). 2019.
- Learning Latent Semantic Annotations for Grounding Natural Language to Structured Data. **G. Qin**, J.-G. Yao, X. Wang, J. Wang, and C.-Y. Lin. In *EMNLP* (oral presentation). 2018.

* *indicates equal contribution;* † *indicates correspondence.* *Last updated on October 14, 2026.*